

Alexander Reinthal

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I am an independent researcher studying how models become subtly misaligned during training. I bring 7 years of industry experience across cyber-security, data, and ML engineering, where I studied IT security failures using machine learning and data.

Research Projects

Testing Scheming Drivers on Instrumental Convergence Evals

May 2026 – present

What happens to instrumental convergence when you change the prompt?

- Found that pressure prompts like ‘achieve your goal in the long run’ have opposite effects on misalignment rates in Gemini 3.1 Pro depending on whether the model is evaluation-aware.

Detecting Deception in Chinese Models

Jan 2026 – present

ARENA 7.0 weekend hackathon, 1st place. Linear probes for deception and politically sensitive lying across frontier open-weight models.

- Linear probes for deception transfer to Llama-3.3-70B (AUROC > 0.74) but fail on Qwen-2.5-72B (AUROC < 0.5) while Qwen is more China-aligned on evals than Llama-3.3. A curious gap that warrants further research. Ongoing.
- Awarded a Bluedot Impact rapid grant to extend the work to additional frontier open-weight models.

About Emergent Misalignment

Jan 2026 – present

ARENA 7.0 capstone, 3rd place. Investigating how emergent misalignment trades off against model capabilities.

- Found that emergent misalignment degrades capabilities by 68pp on GSM8K and that any correct-data SFT simultaneously restores capabilities and reduces EM — effects are inseparable under this training recipe.

Detecting Piecewise Cyber Espionage in Model APIs

Nov 2025 – Nov 2025

Apart Research sprint, 4th of 641 submissions — proof of concept for detecting malicious activity across multiple agent sessions.

- Individual requests can look benign in isolation (100% to 23% detection by LlamaGuard 3 on IP scans) — but can be detected by modelling activity across user sessions.
- Co-authored with Arthur Colle, David Williams-King, Yingquan Li, and Lihn Le.

Experience

AI Safety Researcher

May 2026 – present

Independent

- Researching how subtle misalignment develops during training. Inspired by seminars on RL, inoculation prompting, and natural emergent misalignment, and by my ARENA capstone.

Mentor

Mar 2026 – present

Bluedot Impact

- Part-time facilitator and mentor for the Technical AI Safety course and project.

Alum

Jan 2026 – Feb 2026

ARENA 7.0

- Five-week intensive curriculum on transformer internals, mechanistic interpretability, RLHF, evals, JAX, and the Inspect framework.

Data Platform Engineer / Tech Lead / Data Scientist
Knowit Solutions Cocreate

Gothenburg, Sweden
Feb 2022 – present

- Implemented an OAuth-authenticated MCP gateway to secure agentic workflows and pioneered LLM evals at the company.

Security Analyst
NTT Security

Gothenburg, Sweden
Apr 2019 – Jan 2022

- Built ML classifiers and anomaly-detection pipelines for malicious network traffic across high-volume customer environments using Python and distributed processing.

Python Software Engineer
Ericsson

Gothenburg, Sweden
June 2017 – Oct 2018

- Led a team building a log-parsing and analytics tool that grew into a dedicated Ericsson engineering team.

Publications

Data Modeling for Predicting Software Exploits

Nov 2018

Peer-reviewed paper for predicting which CVEs become exploited in the wild. Adopted by Recorded Future's Vulnerability Intelligence product (threat-intelligence company). Customers report 86% reduction in unplanned downtime, 11 hours/week saved on triage, and 73% increase in threat visibility.

Reinthal, A., Filippakis, E., Almgren, M.

[10.1007/978-3-030-03638-6_21](https://doi.org/10.1007/978-3-030-03638-6_21) (Springer LNCS: Nordic Conference on Secure IT Systems)

Open Source and Personal Projects

Casually Jailbreaking Gemini 2.5 Flash

2026 – 2026

Jailbroke Gemini 2.5 Flash with a few-shot in-context continuation attack, eliciting harmful recommendations.

- Practical demonstration of adversarial-robustness gaps in a production frontier model — directly relevant to the team's automated red-teaming and multi-turn attack work.

Open Source Contributions

github.com/reinthal

- [Inspect AI PR #3987](#) Fixed Gemini Pro 3.1+Openrouter+Reasoning issue
- [Dagster PR #24188](#) (ML pipeline orchestration) and [DLT Hub PR #594](#) (ingestion reliability)

Skills

Machine Learning & AI: PyTorch, JAX (learned during ARENA 7.0), Einops, XGBoost, SciKit Learn, Jupyter; LLM evals & observability (Inspect, LangChain, Langfuse, Arize Phoenix); linear probes, SFT, automated red-teaming

Programming Languages: Python, JavaScript, SQL, Rust, C/C++, R, Nix, Terraform

Platforms & Tools: Modal, Runpod, Openweights, Weights & Biases, Spark, Docker, Kubernetes, AWS, Azure

Certifications: Bluedot Impact — Technical AI Safety (2026-03); AGI Strategy (2025-10)

Education

Chalmers University of Technology
MS in Engineering Physics

Gothenburg, Sweden
Sept 2016 – June 2018

- Coursework: Statistical Physics, Neural Networks and Machine Learning

Gothenburg University
BS in Computer Science

Gothenburg, Sweden
Sept 2013 – June 2016

- Coursework: Algorithms, Testing/Debugging/Verification, Theoretical Computer Science, Operating Systems, Cryptography, Cyber Security